

Debt Service Coverage and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Debt Service Coverage (DSC), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023a\)](#). A value-weighted long/short trading strategy based on DSC achieves an annualized gross (net) Sharpe ratio of 0.39 (0.37), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 34 (39) bps/month with a t-statistic of 2.49 (2.83), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Cash-based operating profitability, Operating Cash flows to price, Return on assets (qtrly), Net Operating Assets, Operating profitability RD adjusted, Accruals) is 32 bps/month with a t-statistic of 3.19.

1 Introduction

Financial markets continuously process vast amounts of information, yet mounting evidence suggests that prices do not immediately incorporate all available data. The speed and completeness of information diffusion across market participants fundamentally shapes asset pricing dynamics and determines the cross-sectional distribution of expected returns. While traditional finance theory assumes rapid price adjustment to new information, behavioral models recognize that investors face cognitive constraints and information processing costs that slow the incorporation of value-relevant signals into prices (Hong and Stein, 1999; DellaVigna, 2009). This gradual diffusion of information creates predictable patterns in stock returns, as prices slowly converge to fundamental values over time.

Despite extensive research on information frictions, we still lack a complete understanding of which fundamental signals investors systematically underreact to and how these patterns vary across different market segments. Corporate financial health indicators, particularly those related to debt servicing capacity, represent critical information for equity valuation yet may receive insufficient attention from investors focused on more salient metrics like earnings or revenue growth. We examine whether Debt Service Coverage (DSC), a measure of firms' ability to meet debt obligations from operating cash flows, predicts cross-sectional stock returns due to investors' gradual incorporation of this information into prices.

The slow diffusion of information hypothesis provides a compelling framework for understanding why DSC might predict future returns. When information disperses gradually across investors, prices underreact to news, creating momentum in returns as information continues to spread (Hong and Stein, 1999). DSC captures firms' financial stability through the relationship between operating cash flows and debt obligations, yet this metric requires synthesizing information from multiple financial statement items and may not receive immediate attention from investors focused

on headline earnings numbers. The complexity of calculating and interpreting DSC creates natural frictions in information processing, as investors must combine cash flow data with detailed debt service requirements to assess firms' true financial health (Hirshleifer et al., 2009).

Limited investor attention amplifies these information processing delays, particularly for complex accounting signals that require careful analysis. Research demonstrates that investors exhibit limited attention to financial statement information beyond simple summary statistics, leading to systematic underreaction to value-relevant accounting data (Hirshleifer et al., 2004; DellaVigna, 2009). DSC exemplifies such a signal—while critically important for assessing default risk and financial flexibility, it does not appear prominently in earnings announcements or analyst reports. This lack of salience means that information about changes in debt service coverage diffuses slowly through the market, first reaching sophisticated investors who actively monitor detailed financial metrics before gradually spreading to the broader investment community.

The gradual incorporation of DSC information should generate predictable return patterns concentrated where information frictions are most severe. Small firms with low analyst coverage face greater information asymmetries, as fewer market participants actively collect and process information about these securities (Hong and Stein, 1999; Zhang, 2006). Similarly, stocks with low institutional ownership or limited media coverage experience slower information diffusion, as these firms lack the intermediaries who typically accelerate information transmission to retail investors. We therefore predict that DSC's return predictability will be strongest among small, neglected stocks where attention constraints bind most tightly and information travels most slowly through the investor population.

We find strong evidence that DSC predicts cross-sectional stock returns, with a value-weighted long/short trading strategy based on DSC achieving an annualized

gross Sharpe ratio of 0.39 and a monthly average excess return of 37 basis points with a t-statistic of 2.38. The strategy’s performance remains robust after controlling for known risk factors, generating a monthly alpha of 34 basis points relative to the Fama-French six-factor model with a t-statistic of 2.49. These results indicate economically meaningful return predictability that cannot be explained by exposure to standard risk factors including market, size, value, profitability, investment, and momentum.

The DSC strategy exhibits remarkable robustness across alternative portfolio construction methodologies and after accounting for transaction costs. Using equal-weighted portfolios or alternative breakpoint definitions, the strategy consistently generates significant abnormal returns, with twenty-two out of twenty-five specifications producing alpha t-statistics exceeding two. After incorporating transaction costs using high-frequency effective spread estimates, the strategy maintains a net Sharpe ratio of 0.37 and generates a significant net alpha of 39 basis points per month relative to the six-factor model with a t-statistic of 2.83. This net performance ranks in the top 6% of anomalies in the factor zoo, demonstrating that DSC offers genuine improvements to the investable opportunity set.

Consistent with gradual information diffusion, we document substantial variation in DSC’s return predictability across size categories. Among the smallest quintile of stocks, the DSC strategy generates an average return of 77 basis points per month with a t-statistic of 2.89 and a six-factor alpha of 35 basis points. The effect attenuates but remains economically meaningful among larger stocks, with the largest size quintile producing an average return of 25 basis points per month. Controlling for six closely related accounting-based anomalies plus the six Fama-French factors, the DSC strategy retains an alpha of 32 basis points per month with a t-statistic of 3.19, confirming that DSC captures distinct information not reflected in existing profitability and accrual measures.

Our study makes three primary contributions to the asset pricing literature on information diffusion and market efficiency. First, we identify DSC as a novel and robust predictor of cross-sectional returns that ranks in the top quintile of known anomalies in terms of Sharpe ratio and net alpha. Unlike many recently discovered anomalies that fail to survive transaction costs or lack economic significance, DSC generates substantial net returns even after accounting for trading costs and controlling for the factor zoo. This finding extends the literature on accounting-based return predictors beyond traditional profitability and accrual measures examined in [Sloan \(1996\)](#), [Novy-Marx \(2013\)](#), and [Ball et al. \(2015\)](#), demonstrating that cash flow-based solvency metrics contain incremental information for equity valuation.

Second, we provide new evidence supporting gradual information diffusion as a mechanism driving cross-sectional return predictability. While [Hong and Stein \(1999\)](#) document momentum profits arising from slow information diffusion and [DellaVigna \(2009\)](#) review evidence of investor inattention, we show that complex accounting signals like DSC generate return predictability through similar channels. The concentration of DSC’s predictive power among small, low-attention stocks aligns with theoretical predictions about where information frictions should be most binding. Our results complement recent work by [Engelberg et al. \(2018\)](#) on anomaly return concentration in hard-to-arbitrage stocks while highlighting the specific role of information complexity in generating these patterns.

Third, our comprehensive analysis using the [Novy-Marx and Velikov \(2023b\)](#) protocol establishes new standards for anomaly documentation and demonstrates DSC’s incremental contribution to the investment opportunity set. The DSC signal improves the mean-variance efficient frontier even when combined with 158 existing anomalies, suggesting that it captures genuinely new information about expected returns. These findings have important implications for both academic research on market efficiency and practical portfolio construction, as they identify an imple-

mentable strategy that expands the feasible investment opportunity set after accounting for real-world frictions. The persistence of DSC’s predictive power suggests that information processing constraints create lasting impediments to market efficiency, even in the presence of sophisticated investors and advanced information technology.

2 Data

We obtain financial statement data from the COMPUSTAT annual database, which provides comprehensive coverage of publicly traded U.S. companies. Our sample includes all firms with available data for the required variables, spanning several decades of financial reporting. We focus on annual observations to ensure consistency in the measurement of cash flows and debt obligations across firms. Debt Service Coverage signal requires two key variables from COMPUSTAT. Operating activities net cash flow (OANCF) represents the cash generated from a firm’s core business operations, calculated as net income adjusted for non-cash charges and changes in working capital. This measure captures the actual cash available to the firm after accounting for the cash requirements of day-to-day operations. Debt in current liabilities (DLC) represents the portion of long-term debt that is due within one year, including the current portion of long-term debt, notes payable, and other short-term borrowings that must be repaid or refinanced within the current operating cycle. We construct the Debt Service Coverage signal by dividing operating activities net cash flow (OANCF) by debt in current liabilities (DLC). This ratio measures a firm’s ability to service its near-term debt obligations using cash generated from operations, providing insight into the company’s short-term financial health and liquidity position. Higher values indicate stronger capacity to meet immediate debt payments without requiring external financing or asset liquidation. We calculate this signal using fiscal year-end values and require non-missing values for both OANCF

and DLC. To address potential outliers and ensure economic meaningfulness, we exclude observations where DLC equals zero, as these would result in undefined ratios, and winsorize the resulting signal at the 1st and 99th percentiles.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the DSC signal. Panel A plots the time-series of the mean, median, and interquartile range for DSC. On average, the cross-sectional mean (median) DSC is 52.15 (1.51) over the 1988 to 2024 sample, where the starting date is determined by the availability of the input DSC data. The signal’s interquartile range spans -5.33 to 13.11. Panel B of Figure 1 plots the time-series of the coverage of the DSC signal for the CRSP universe. On average, the DSC signal is available for 5.73% of CRSP names, which on average make up 7.06% of total market capitalization.

4 Does DSC predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on DSC using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high DSC portfolio and sells the low DSC portfolio. The rest of Panel A reports the portfolios’ monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short DSC strategy earns an average return of 0.37% per month with a t-statistic of 2.38. The annualized Sharpe ratio of the strategy is 0.39. The alphas range from 0.34% to 0.53% per

month and have t-statistics exceeding 2.49 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the [Fama and French \(2018\)](#) six-factor model. The long/short strategy's most significant loading is -0.68, with a t-statistic of -11.53 on the HML factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 456 stocks and an average market capitalization of at least \$2,160 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using cap breakpoints and value-weighted portfolios, and equals 32 bps/month with a t-statistics of 2.28. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-two exceed two, and for twelve exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas

measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between 30-60bps/month. The lowest return, (30 bps/month), is achieved from the quintile sort using cap breakpoints and value-weighted portfolios, and has an associated t-statistic of 2.15. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the DSC trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-three cases, and significantly expands the achievable frontier in twenty-two cases.

Table 3 provides direct tests for the role size plays in the DSC strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and DSC, as well as average returns and alphas for long/short trading DSC strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the DSC strategy achieves an average return of 25 bps/month with a t-statistic of 1.47. Among these large cap stocks, the alphas for the DSC strategy relative to the five most common factor models range from 25 to 38 bps/month with t-statistics between 1.44 and 2.80.

5 How does DSC perform relative to the zoo?

Figure 2 puts the performance of DSC in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212

documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the DSC strategy falls in the distribution. The DSC strategy’s gross (net) Sharpe ratio of 0.39 (0.37) is greater than 82% (94%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the DSC strategy (red line).² Ignoring trading costs, a \$1 invested in the DSC strategy would have yielded \$3.49 which ranks the DSC strategy in the top 2% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the DSC strategy would have yielded \$3.08 which ranks the DSC strategy in the top 1% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the DSC relative to those. Panel A shows that the DSC strategy gross alphas fall between the 75 and 90 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 198806 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The DSC strategy has a positive net generalized alpha for five out of the five factor models. In these cases DSC ranks between the 91 and 98 percentiles in terms of how much it could have expanded the achievable investment frontier.

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

6 Does DSC add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of DSC with 209 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price DSC or at least to weaken the power DSC has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of DSC conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{DSC} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{DSC}DSC_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{DSC,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on DSC. Stocks are finally grouped into five DSC portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

DSC trading strategies conditioned on each of the 209 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on DSC and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the DSC signal in these Fama-MacBeth regressions exceed -0.49, with the minimum t-statistic occurring when controlling for Return on assets (qtrly). Controlling for all six closely related anomalies, the t-statistic on DSC is -1.35.

Similarly, Table 5 reports results from spanning tests that regress returns to the DSC strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the DSC strategy earns alphas that range from 24-59bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 1.86, which is achieved when controlling for Return on assets (qtrly). Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the DSC trading strategy achieves an alpha of 32bps/month with a t-statistic of 3.19.

7 Does DSC add relative to the whole zoo?

Finally, we can ask how much adding DSC to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 158 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 158 anomalies augmented with the DSC signal.⁴ We consider

⁴We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capital-

one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 158-anomaly combination strategy grows to \$36.86, while \$1 investment in the combination strategy that includes DSC grows to \$42.26.

8 Conclusion

This study provides compelling evidence that Debt Service Coverage (DSC) represents a robust and economically significant predictor of cross-sectional stock returns. Our analysis, following the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that DSC-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for well-established risk factors and closely related accounting-based signals.

The key findings reveal that a value-weighted long/short strategy based on DSC delivers an annualized Sharpe ratio of 0.39 gross (0.37 net of transaction costs), indicating strong economic significance even after accounting for implementation frictions. Most notably, the strategy’s ability to generate significant alpha relative to the Fama-French five-factor model plus momentum (34-39 bps monthly) suggests that DSC captures a distinct dimension of firm fundamentals not fully explained by existing factor models. The persistence of abnormal returns (32 bps monthly, t -statistic = 3.19) even after controlling for six closely related accounting-based strategies—including cash-based operating profitability, operating cash flows to price, and accruals—underscores DSC’s unique information content in predicting future returns.

ization on CRSP in the period for which DSC is available.

From a practical perspective, these results suggest that DSC warrants serious consideration for inclusion in quantitative investment strategies and risk models. The signal’s robustness to transaction costs and its orthogonality to existing factors make it particularly attractive for portfolio construction and risk management applications. The economic magnitude of the returns, combined with statistical significance, indicates that DSC captures meaningful variation in firms’ financial health that is not fully reflected in current market prices.

However, this study is subject to several limitations that merit acknowledgment. First, while we control for numerous related factors, the possibility remains that DSC proxies for an omitted risk factor rather than representing a market inefficiency. Second, our analysis is confined to U.S. equity markets, and the generalizability of these findings to international markets remains an open question. Third, the implementation of DSC-based strategies may face practical challenges in terms of data availability and calculation complexity that could affect real-world performance.

Future research should explore several promising avenues. First, investigating the economic mechanisms underlying DSC’s predictive power would enhance our understanding of why this signal generates abnormal returns. Second, examining DSC’s performance across different market conditions, particularly during financial crises when debt servicing becomes critical, could provide insights into its risk characteristics. Third, extending the analysis to international markets would test the robustness of these findings across different institutional and regulatory environments. Finally, exploring potential interactions between DSC and other firm characteristics, such as credit ratings or leverage ratios, might reveal conditional effects that could further improve return prediction. Understanding whether DSC’s predictive power has varied over time and what drives such variation would also contribute to the broader debate about factor decay and the evolution of market efficiency.

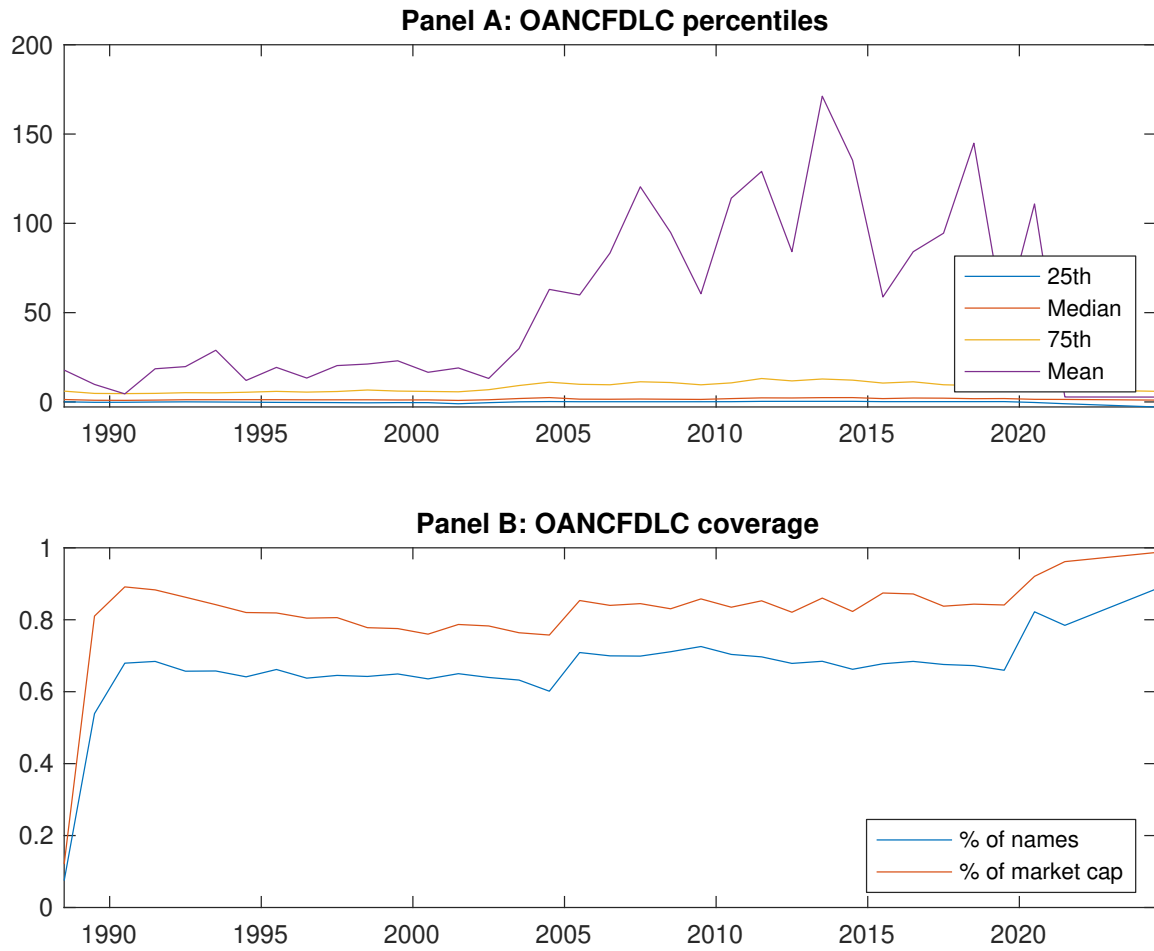


Figure 1: Times series of DSC percentiles and coverage.
This figure plots descriptive statistics for DSC. Panel A shows cross-sectional percentiles of DSC over the sample. Panel B plots the monthly coverage of DSC relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on DSC. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 198806 to 202406.

Panel A: Excess returns and alphas on DSC-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.48 [1.72]	0.67 [3.52]	0.83 [4.40]	0.74 [3.82]	0.85 [3.39]	0.37 [2.38]
α_{CAPM}	-0.42 [-3.68]	0.04 [0.61]	0.21 [2.94]	0.09 [1.43]	0.01 [0.10]	0.43 [2.75]
α_{FF3}	-0.48 [-4.83]	-0.00 [-0.03]	0.18 [2.76]	0.08 [1.26]	0.05 [0.73]	0.53 [4.00]
α_{FF4}	-0.41 [-4.13]	0.02 [0.26]	0.17 [2.55]	0.07 [1.06]	0.08 [1.11]	0.49 [3.65]
α_{FF5}	-0.23 [-2.54]	-0.13 [-2.10]	0.02 [0.27]	-0.06 [-0.94]	0.13 [1.70]	0.36 [2.69]
α_{FF6}	-0.19 [-2.07]	-0.10 [-1.71]	0.02 [0.27]	-0.06 [-0.97]	0.15 [1.94]	0.34 [2.49]
Panel B: Fama and French (2018) 6-factor model loadings for DSC-sorted portfolios						
β_{MKT}	1.12 [48.79]	0.92 [61.04]	0.92 [59.06]	0.94 [62.19]	1.07 [57.27]	-0.05 [-1.38]
β_{SMB}	0.06 [1.69]	-0.04 [-1.81]	-0.08 [-3.48]	0.01 [0.52]	0.05 [1.91]	-0.00 [-0.09]
β_{HML}	0.53 [13.22]	0.05 [1.89]	-0.06 [-2.22]	-0.07 [-2.73]	-0.15 [-4.52]	-0.68 [-11.53]
β_{RMW}	-0.38 [-9.07]	0.18 [6.70]	0.26 [9.27]	0.24 [8.60]	-0.11 [-3.14]	0.27 [4.44]
β_{CMA}	-0.30 [-5.09]	0.22 [5.69]	0.22 [5.61]	0.14 [3.61]	-0.10 [-2.09]	0.20 [2.30]
β_{UMD}	-0.07 [-3.32]	-0.04 [-2.77]	-0.00 [-0.03]	0.00 [0.26]	-0.03 [-1.81]	0.04 [1.26]
Panel C: Average number of firms (n) and market capitalization (me)						
n	1227	501	456	486	503	
me (\$10 ⁶)	2160	2758	3786	3274	2697	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the DSC strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 198806 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.37 [2.38]	0.43 [2.75]	0.53 [4.00]	0.49 [3.65]	0.36 [2.69]	0.34 [2.49]
Quintile	NYSE	EW	0.54 [2.91]	0.59 [3.19]	0.50 [3.03]	0.38 [2.29]	0.09 [0.64]	0.00 [0.03]
Quintile	Name	VW	0.64 [2.80]	0.96 [4.45]	0.88 [4.93]	0.77 [4.29]	0.35 [2.27]	0.28 [1.81]
Quintile	Cap	VW	0.32 [2.28]	0.38 [2.65]	0.47 [3.86]	0.41 [3.40]	0.34 [2.80]	0.30 [2.47]
Decile	NYSE	VW	0.60 [2.90]	0.80 [3.91]	0.82 [4.53]	0.77 [4.19]	0.47 [2.68]	0.45 [2.52]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.35 [2.24]	0.43 [2.76]	0.51 [3.88]	0.49 [3.66]	0.40 [3.04]	0.39 [2.83]
Quintile	NYSE	EW	0.38 [2.01]	0.44 [2.28]	0.35 [2.06]	0.28 [1.65]		
Quintile	Name	VW	0.60 [2.61]	0.91 [4.23]	0.82 [4.56]	0.75 [4.20]	0.36 [2.37]	0.31 [2.07]
Quintile	Cap	VW	0.30 [2.15]	0.37 [2.57]	0.44 [3.62]	0.41 [3.35]	0.36 [2.94]	0.33 [2.68]
Decile	NYSE	VW	0.56 [2.72]	0.77 [3.75]	0.78 [4.22]	0.74 [4.04]	0.50 [2.84]	0.48 [2.72]

Table 3: Conditional sort on size and DSC

This table presents results for conditional double sorts on size and DSC. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on DSC. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high DSC and short stocks with low DSC. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 198806 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	DSC Quintiles					DSC Strategies						
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
	(1)	0.26 [0.58]	0.26 [0.75]	0.56 [1.92]	1.05 [3.50]	1.03 [3.59]	0.77 [2.89]	1.02 [3.91]	0.82 [3.96]	0.79 [3.71]	0.38 [2.10]	0.35 [1.94]
	(2)	0.16 [0.39]	0.69 [2.48]	0.92 [3.31]	0.76 [2.80]	0.88 [3.11]	0.72 [3.07]	0.97 [4.29]	0.82 [4.25]	0.77 [3.94]	0.28 [1.71]	0.26 [1.57]
	(3)	0.23 [0.70]	0.87 [3.52]	1.00 [3.82]	0.73 [2.76]	0.82 [3.01]	0.59 [3.37]	0.76 [4.43]	0.71 [4.22]	0.64 [3.78]	0.31 [1.98]	0.27 [1.74]
	(4)	0.55 [1.89]	0.81 [3.66]	0.86 [3.64]	0.80 [3.20]	0.81 [3.18]	0.27 [1.82]	0.36 [2.50]	0.34 [2.36]	0.29 [2.01]	-0.01 [-0.04]	-0.03 [-0.22]
	(5)	0.57 [2.21]	0.63 [3.40]	0.88 [4.60]	0.71 [3.75]	0.82 [3.20]	0.25 [1.47]	0.25 [1.44]	0.38 [2.80]	0.36 [2.57]	0.33 [2.33]	0.31 [2.18]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	DSC Quintiles					DSC Quintiles						
		Average n					Average market capitalization (\$10 ⁶)					
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)	
	(1)	337	335	342	346	347	32	28	34	43	52	
	(2)	101	101	101	101	101	66	71	73	71	73	
	(3)	71	71	72	71	72	123	130	129	128	130	
	(4)	62	62	62	62	62	276	298	288	281	285	
(5)	59	59	58	59	59	2197	2210	3043	2576	2040		

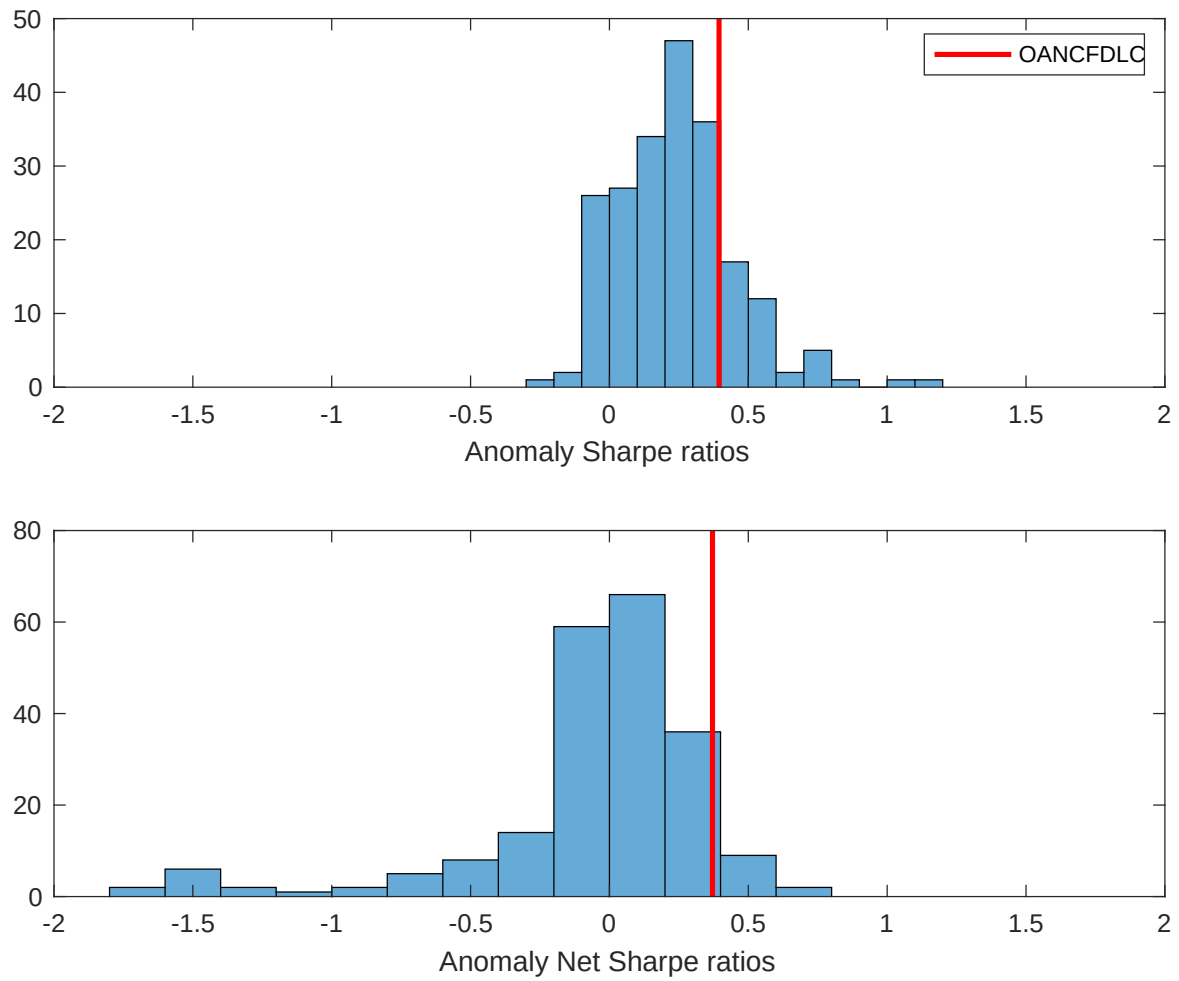


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the DSC with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

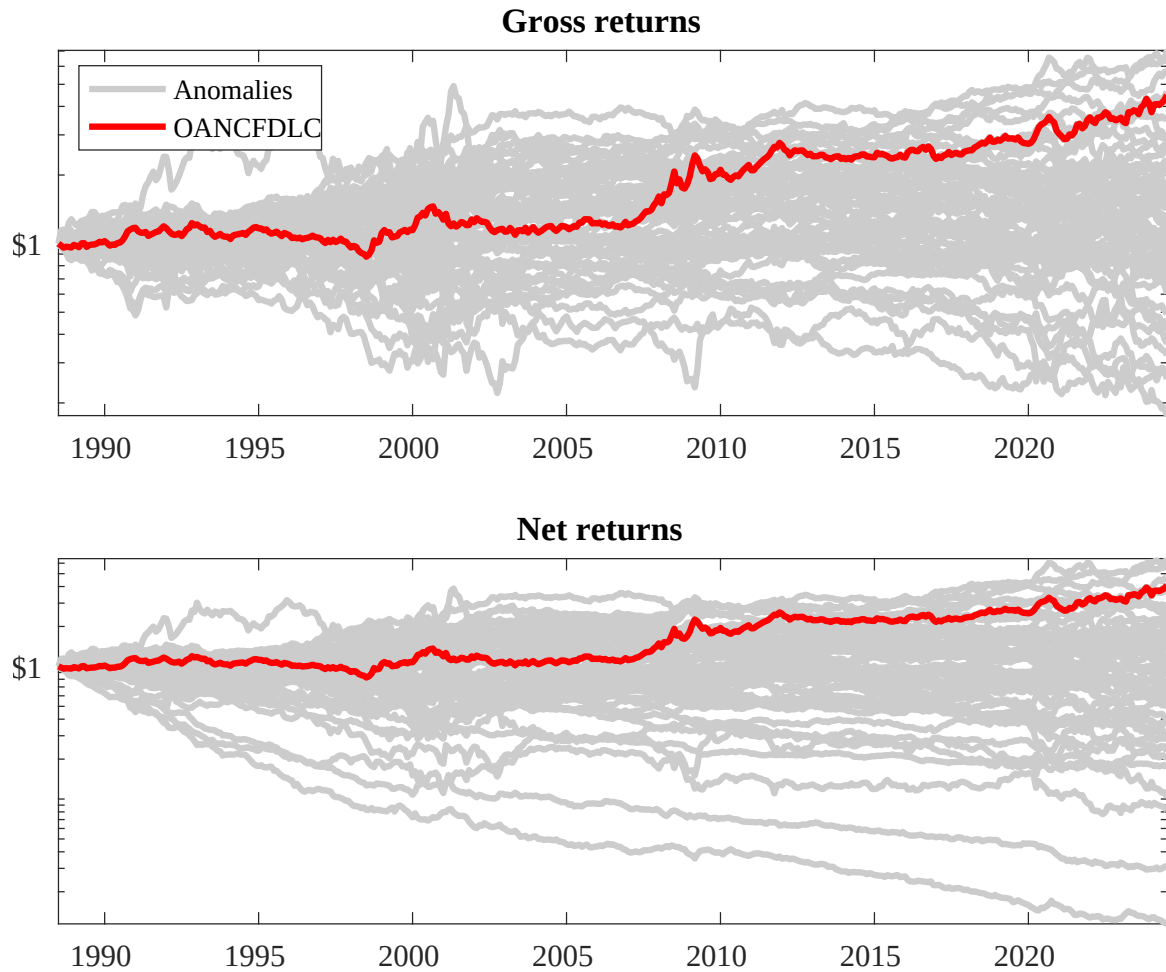
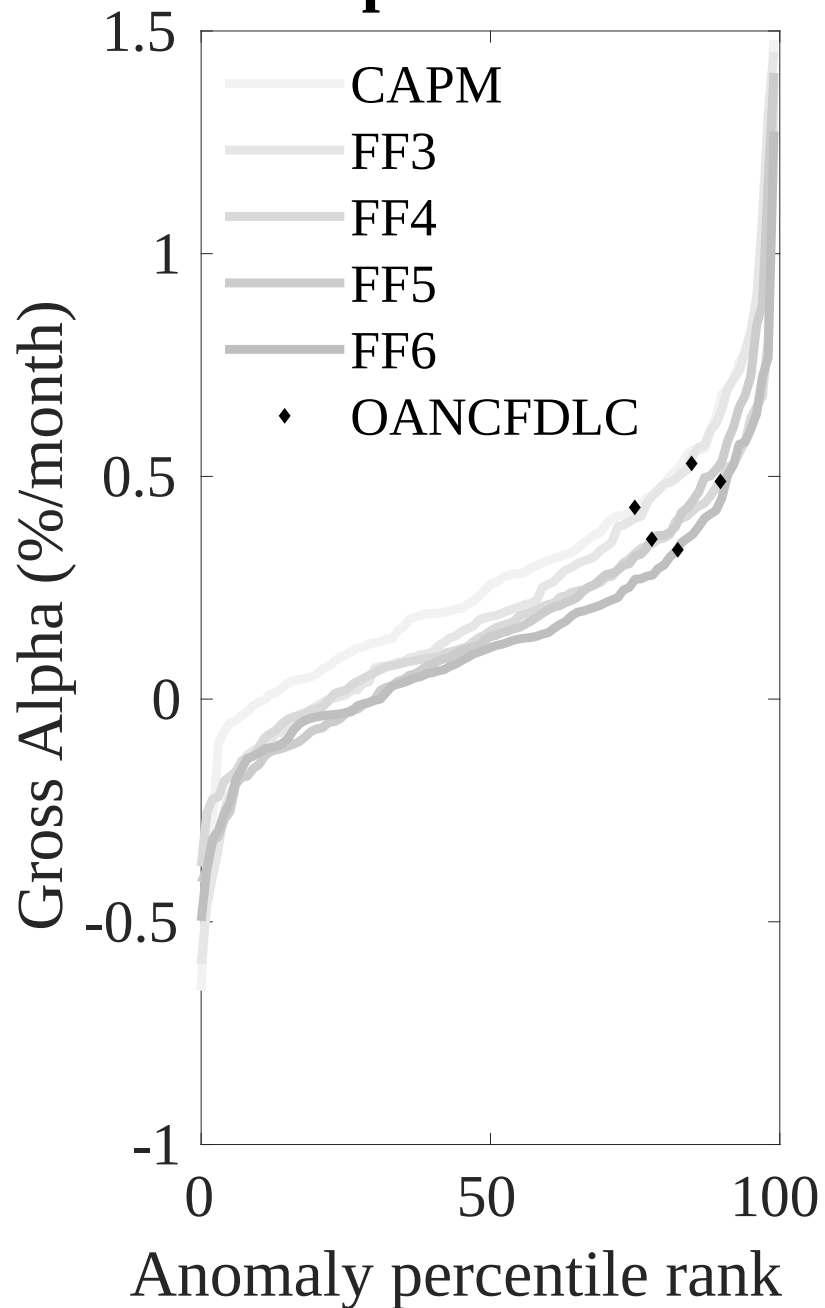


Figure 3: Dollar invested.

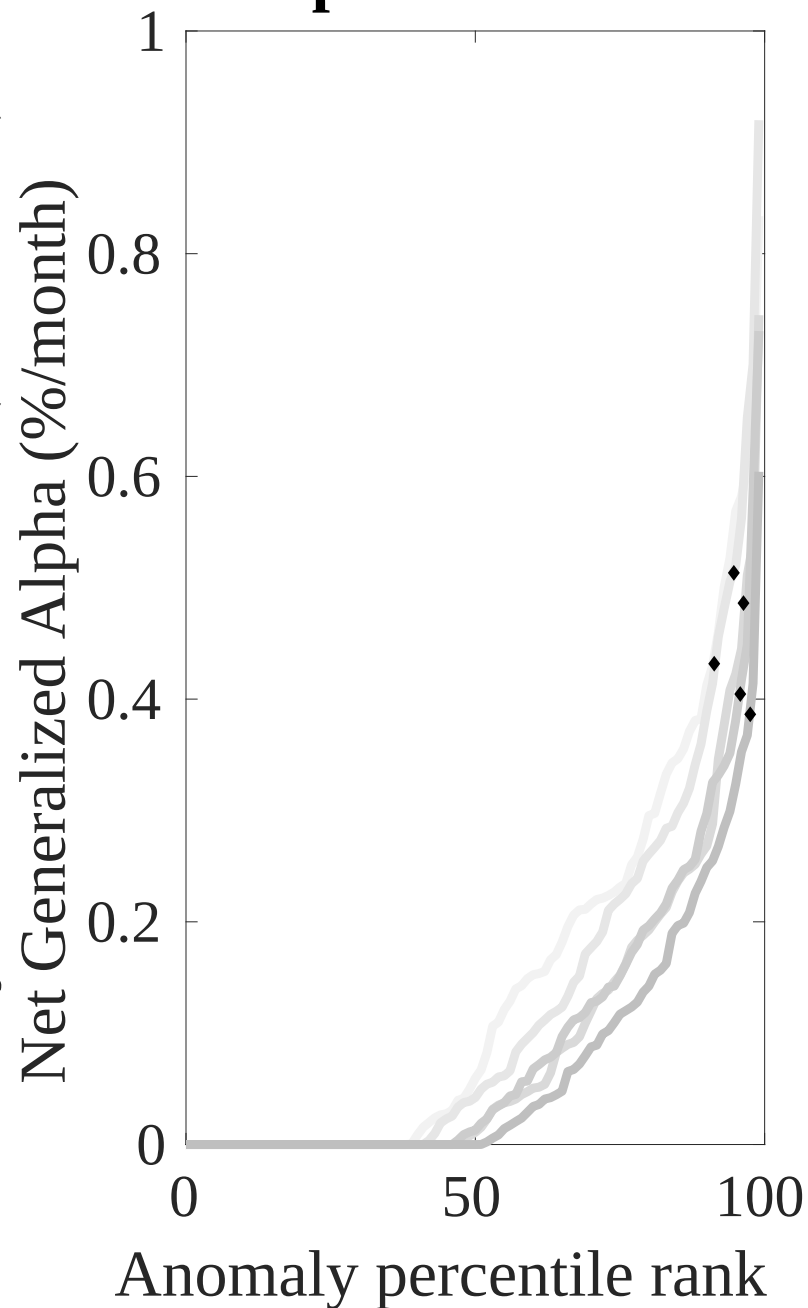
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the DSC trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

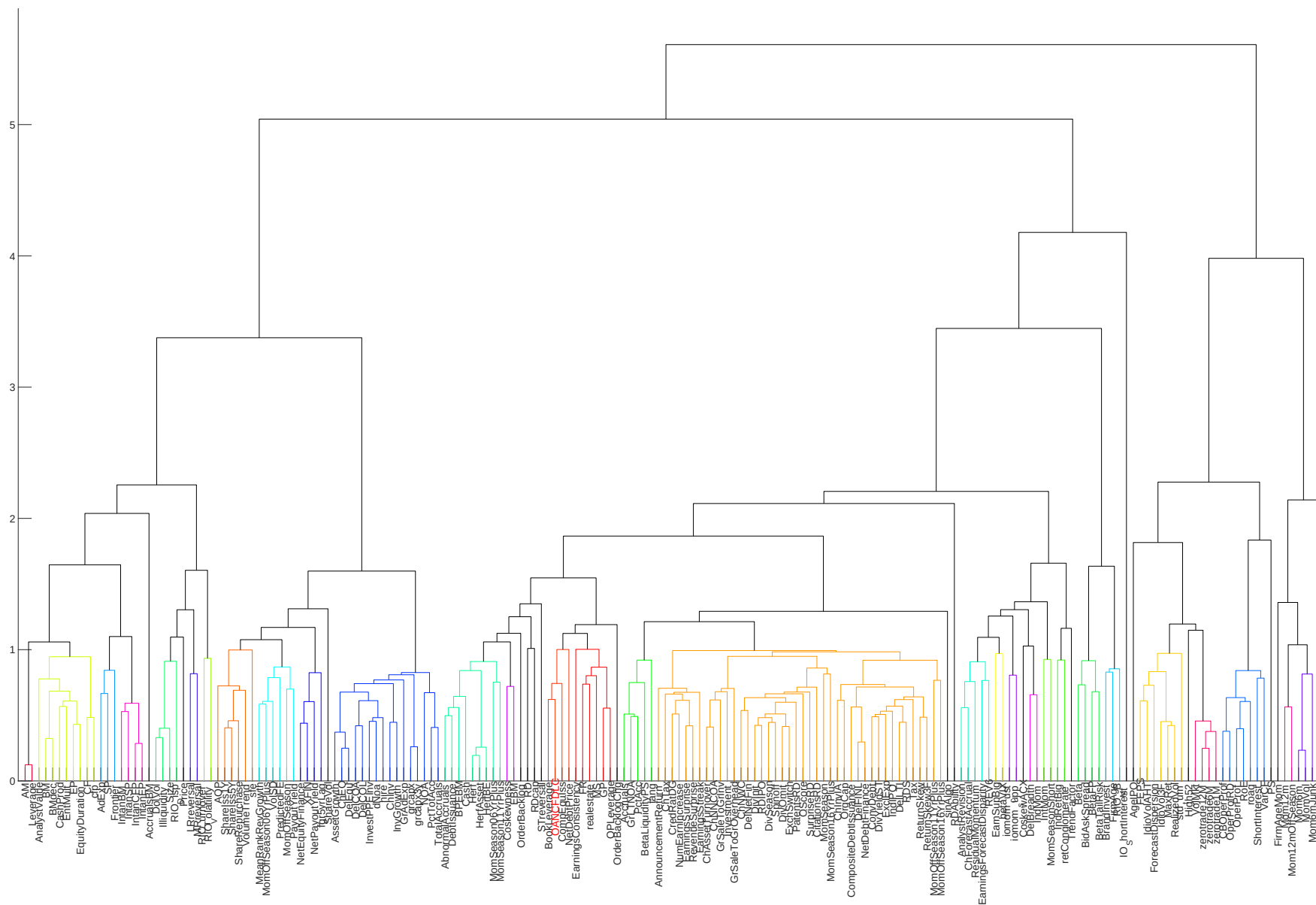
Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

Net Alpha Percentiles





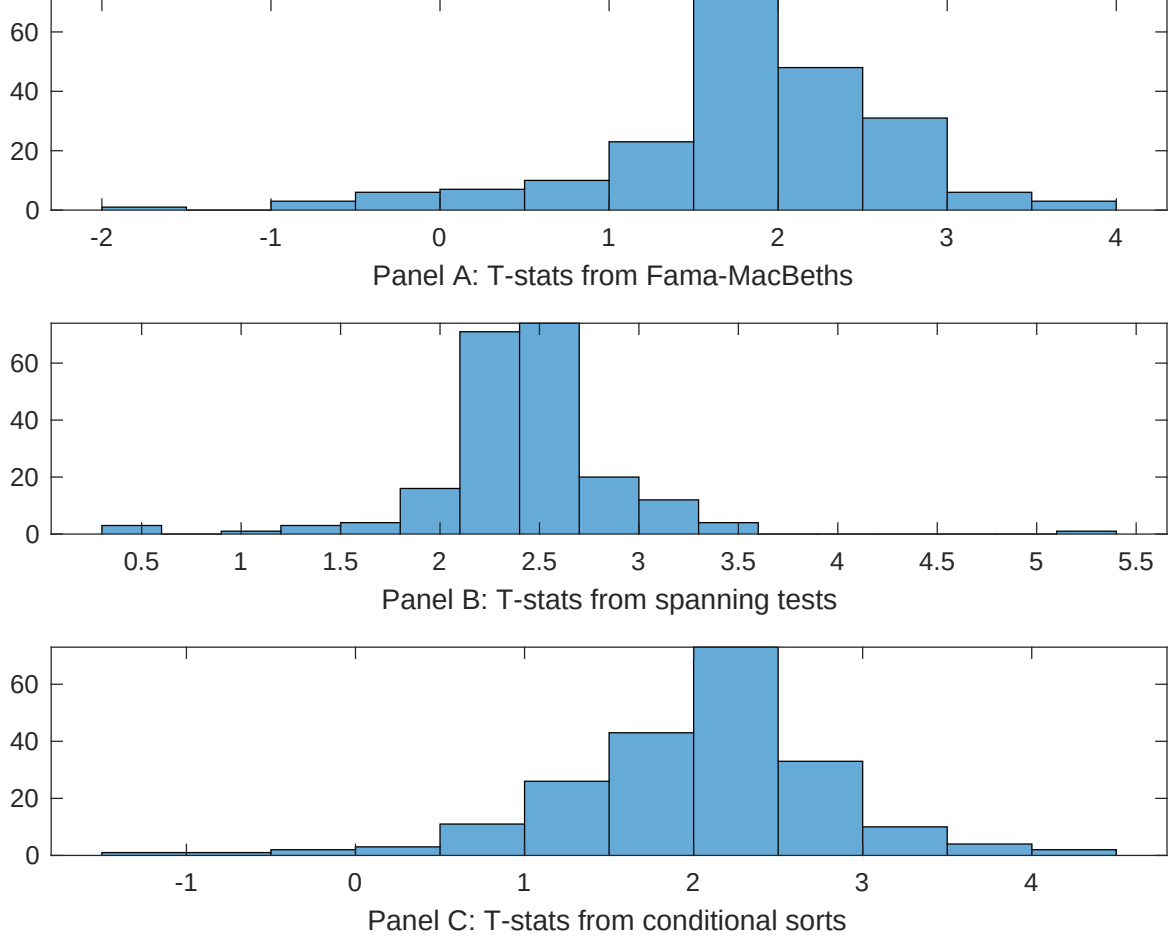


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of DSC conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{DSC} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{DSC} DSC_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{DSC,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on DSC. Stocks are finally grouped into five DSC portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted DSC trading strategies conditioned on each of the 209 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on DSC. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{DSC}DSC_{i,t} + \sum_{k=1}^s \beta_{X_k}X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Cash-based operating profitability, Operating Cash flows to price, Return on assets (qtrly), Net Operating Assets, Operating profitability RD adjusted, Accruals. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 198806 to 202406.

Intercept	0.90 [2.83]	0.95 [3.26]	0.11 [3.96]	0.14 [4.58]	0.85 [2.54]	1.00 [3.61]	0.15 [4.85]
DSC	0.24 [0.05]	0.71 [1.42]	-0.18 [-0.49]	0.13 [2.04]	0.20 [0.43]	0.10 [1.53]	-0.46 [-1.35]
Anomaly 1	0.21 [4.70]						-0.70 [-1.00]
Anomaly 2		0.10 [3.88]					0.46 [1.63]
Anomaly 3			0.49 [2.84]				0.45 [2.81]
Anomaly 4				0.53 [3.76]			0.87 [7.79]
Anomaly 5					0.20 [3.76]		0.18 [2.63]
Anomaly 6						0.13 [3.09]	0.17 [2.79]
# months	427	427	427	432	427	432	427
$\bar{R}^2(\%)$	1	1	1	0	1	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the DSC trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{DSC} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Cash-based operating profitability, Operating Cash flows to price, Return on assets (qtrly), Net Operating Assets, Operating profitability RD adjusted, Accruals. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 198806 to 202406.

Intercept	0.25 [1.86]	0.32 [2.33]	0.30 [2.25]	0.59 [5.10]	0.29 [2.12]	0.24 [2.08]	0.32 [3.19]
Anomaly 1	27.03 [4.61]						31.01 [4.80]
Anomaly 2		11.55 [2.17]					4.32 [1.10]
Anomaly 3			26.44 [4.47]				15.54 [3.33]
Anomaly 4				-61.30 [-12.59]			-49.69 [-10.80]
Anomaly 5					12.76 [2.28]		-4.57 [-0.75]
Anomaly 6						57.51 [13.39]	37.07 [9.10]
mkt	-2.31 [-0.69]	-4.71 [-1.40]	-0.79 [-0.23]	-1.71 [-0.59]	-2.51 [-0.72]	-3.43 [-1.21]	2.37 [0.93]
smb	7.60 [1.53]	-0.66 [-0.14]	6.37 [1.30]	-15.44 [-3.57]	4.49 [0.89]	8.70 [2.12]	2.02 [0.51]
hml	-55.57 [-8.93]	-74.09 [-10.66]	-57.44 [-9.40]	-44.90 [-8.45]	-60.90 [-9.62]	-45.27 [-8.72]	-23.10 [-4.06]
rmw	12.78 [1.91]	22.58 [3.55]	2.39 [0.30]	21.79 [4.17]	16.97 [2.30]	40.31 [7.72]	3.66 [0.56]
cma	13.42 [1.58]	14.79 [1.69]	17.09 [2.03]	23.71 [3.23]	17.93 [2.09]	-3.24 [-0.44]	1.36 [0.21]
umd	2.03 [0.68]	9.50 [2.48]	-2.48 [-0.75]	8.11 [3.15]	2.63 [0.85]	0.98 [0.39]	1.25 [0.42]
# months	428	428	428	432	428	432	428
$\bar{R}^2(\%)$	36	33	36	51	33	53	66

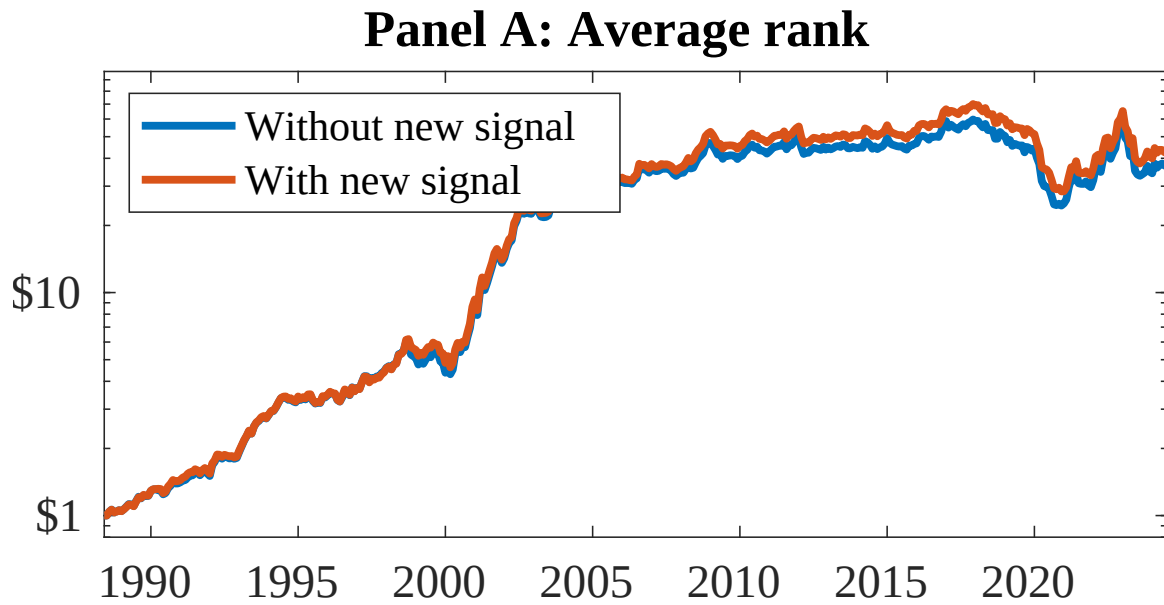


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 158 anomalies. The red solid lines indicate combination trading strategies that utilize the 158 anomalies as well as DSC. Panel A shows results using "Average rank" as the combination method. See [Section 7](#) for details on the combination methods.

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